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EVENT AND DATE

LECTURE 7 – SUBSYSTEM ENGINEERING

CS39A0040 Product & Service Development

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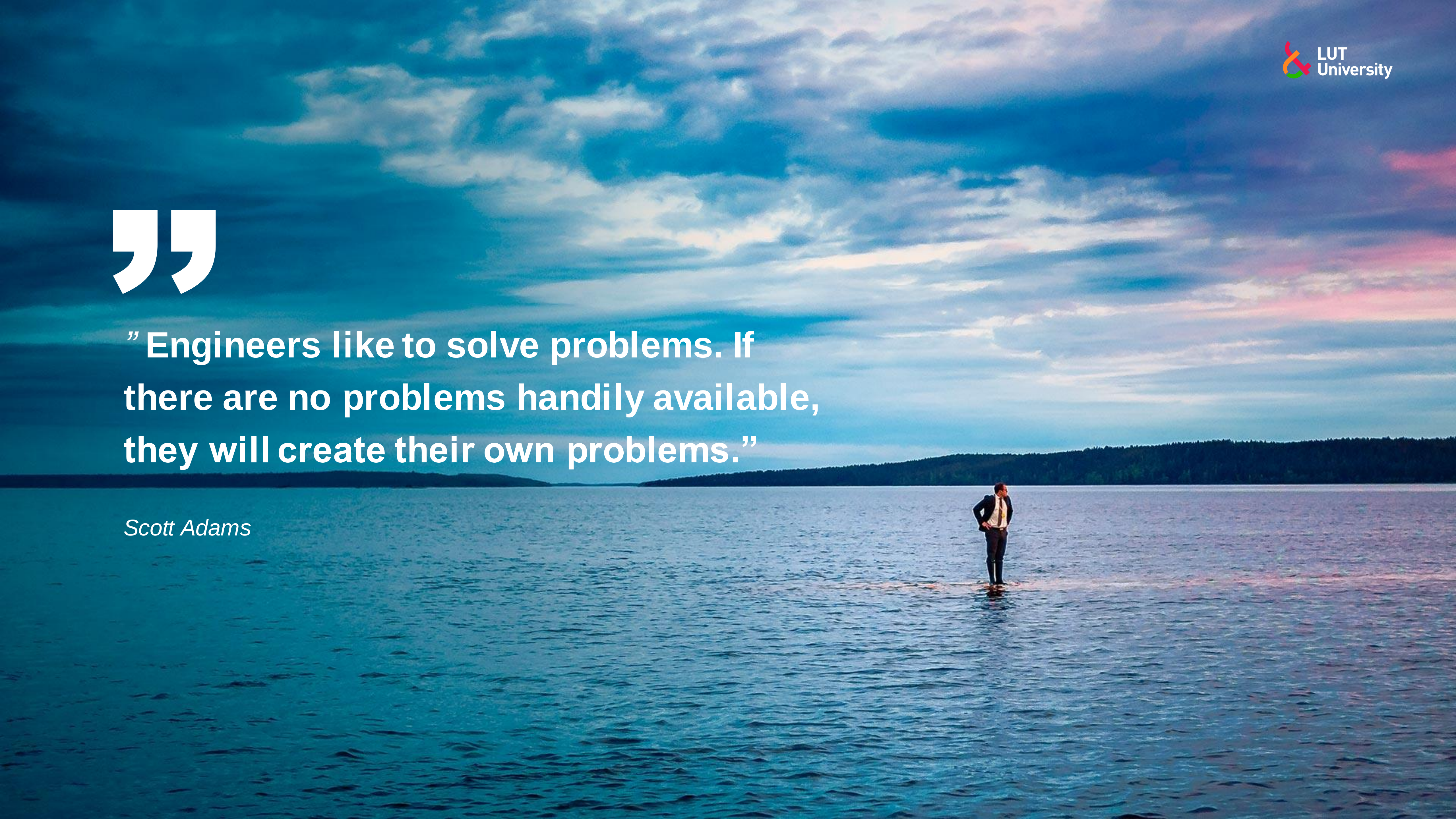
AGENDA FOR LECTURE

- Lecture 7 – Subsystem Engineering
  - Product Management Viewpoint
  - Product Development Team Viewpoint



”Engineers like to solve problems. If there are no problems handily available, they will create their own problems.”

Scott Adams

SUBSYSTEM ENGINEERING - THE BIG PICTURE

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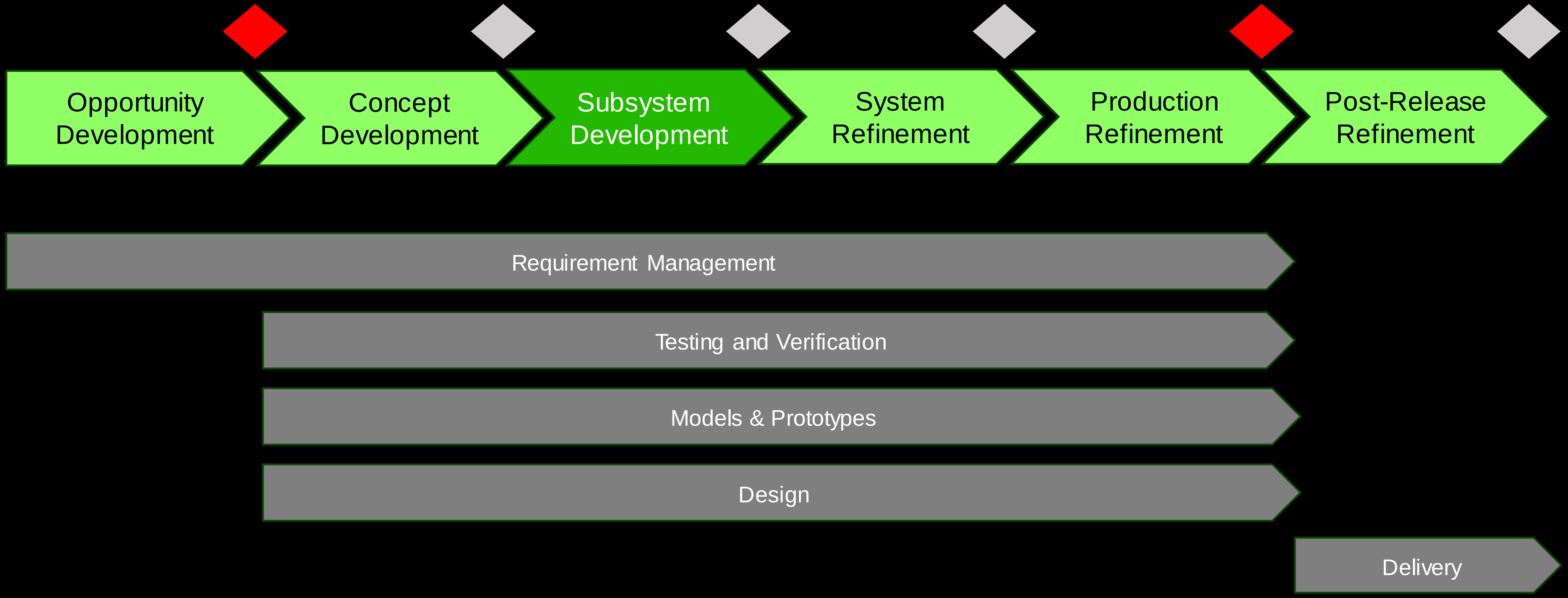
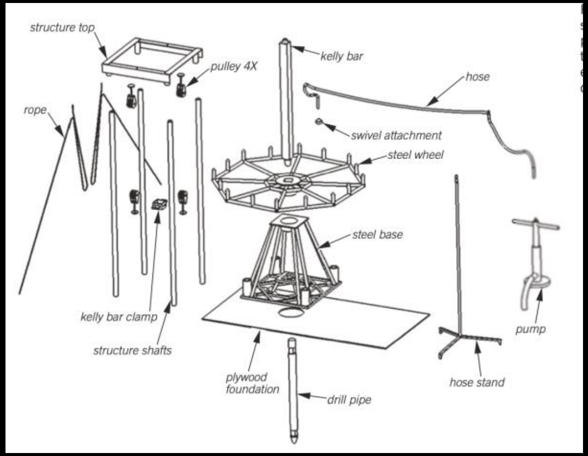
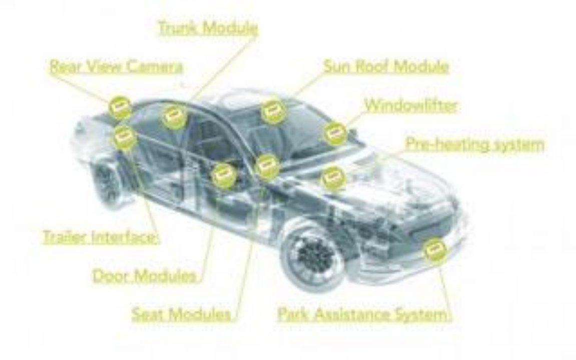
PRODUCT DEVELOPMENT

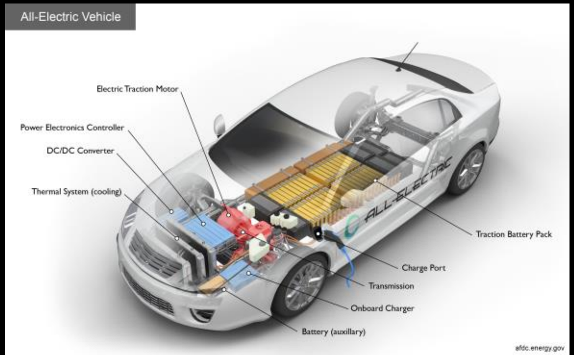


SELECTED CONCEPT : SUBSYSTEMS & INTERFACES

Subsystem Modules







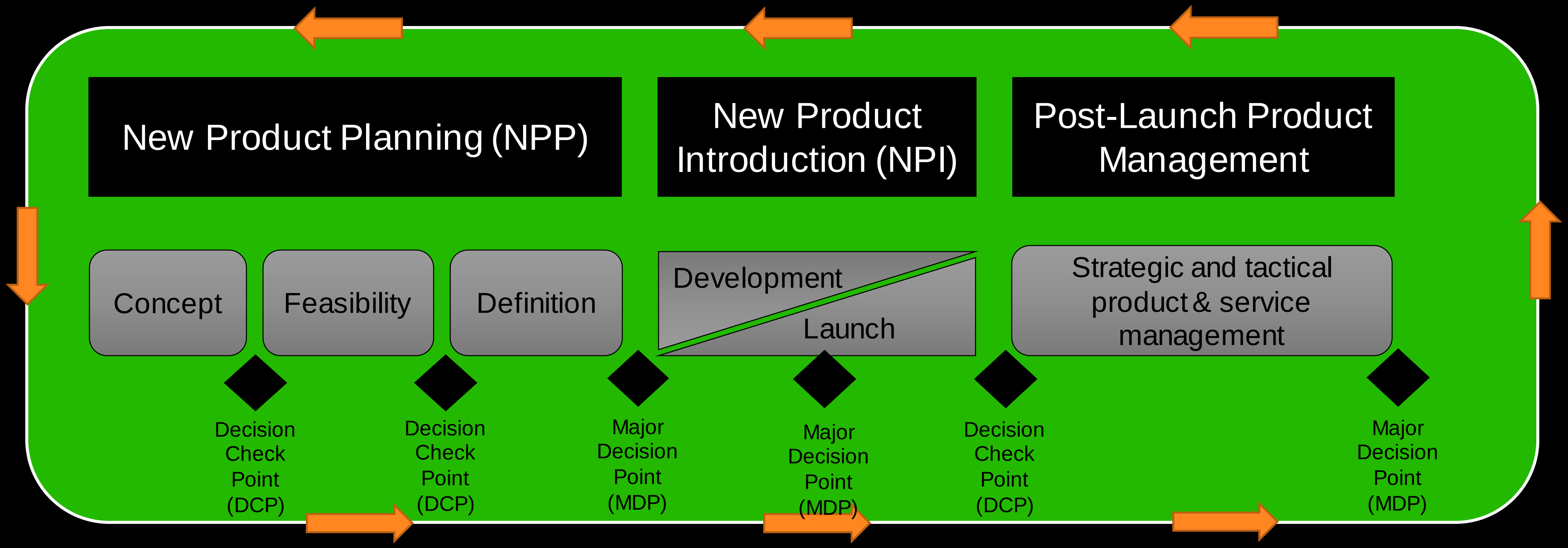
Cocnpet: System & Subsystems

System Interfaces

SUBSYSTEM ENGINEERING PHASE

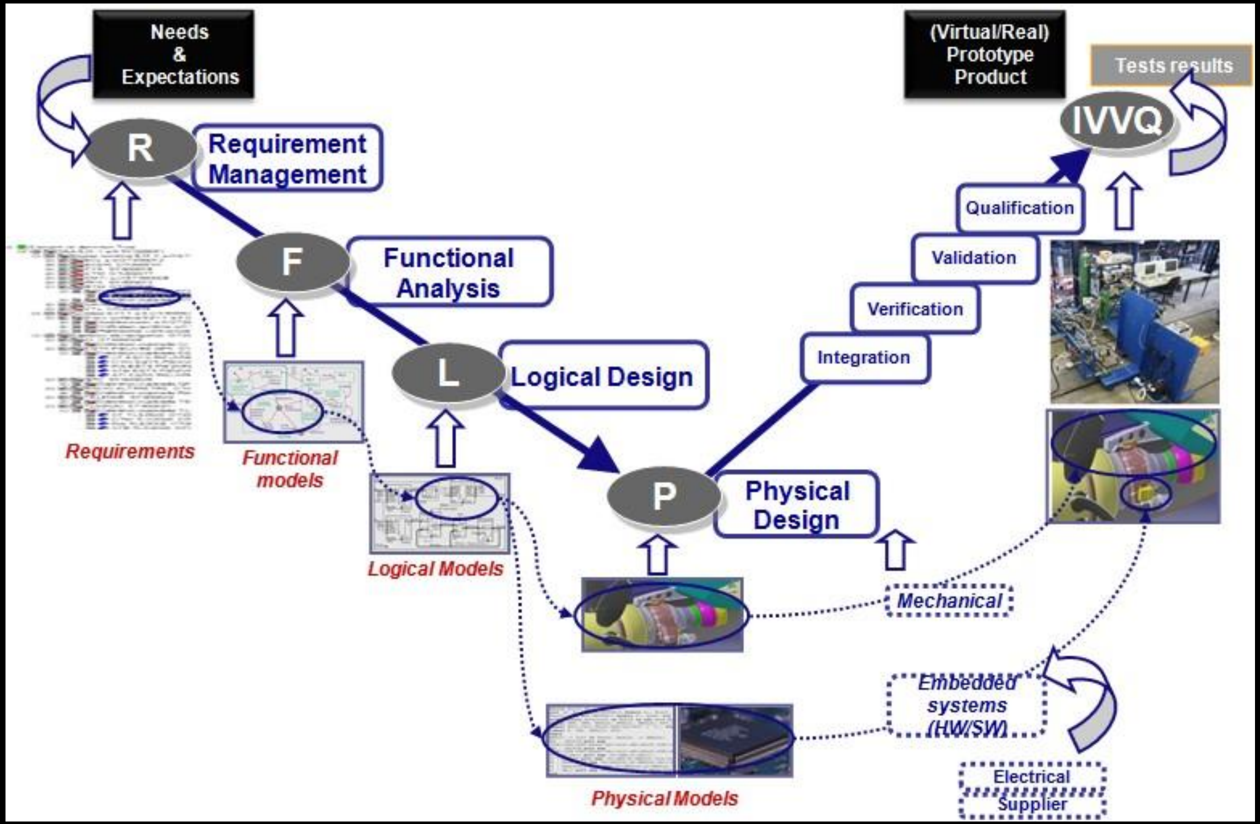
Methods to define subsystems for Product Development & Product Mangement

GENERIC PLANNING & DEVELOPMENT PROCESS



Steven Haines, The Product Manager's Desk Reference

DEVELOPMENT PROCESS EXAMPLES



Systems Engineering Example

DEVELOPMENT PROCESS EXAMPLES

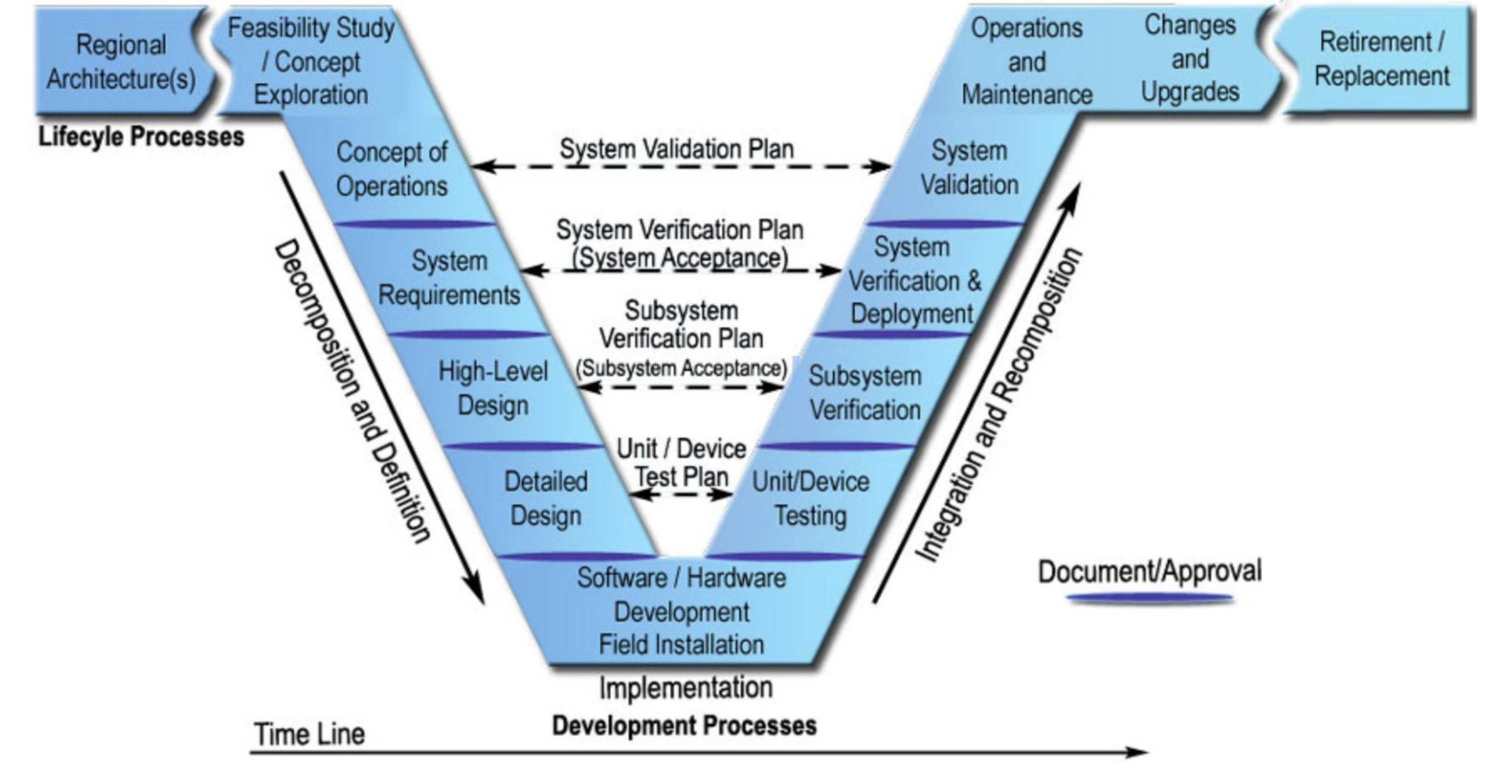
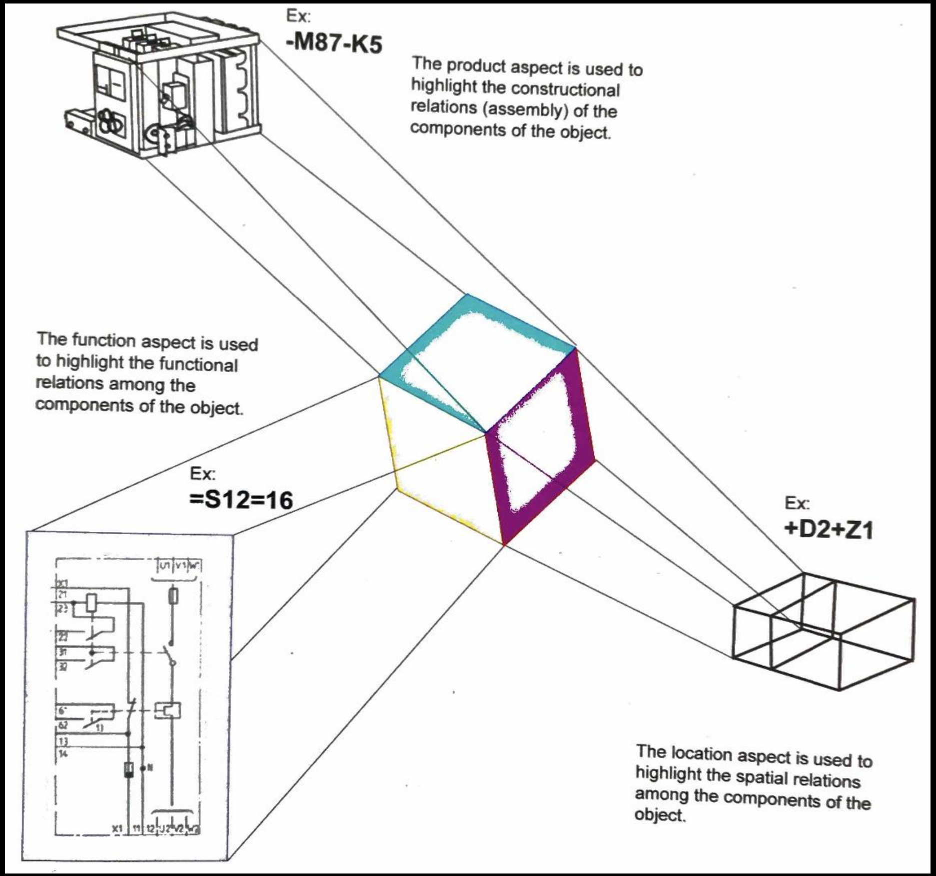


Figure 7: Systems Engineering "V" Diagram

Systems Engineering Example

TYPICAL PRODUCT DEFINITION VIEWS



- The physical aspect is used to highlight the constructional relations of the component of the object.
- The functional aspect is used to highlight the functional relations among the component of the object.
- The location aspect is used to highlight the spatial relations among the components of the object.
- There can be more product & service viewpoints.
- Avoid object overloading.

PRODUCT, SYSTEM & SUBSYSTEM MODULARITY

Variations

Customized products

- Designed and manufactured on order

Modularized products

- Mass customization
- Ready-made designs & manufactured on order

Standardized products

- Mass production
- Designed and manufactured to stock

Production

- Manufacturing industry balances between customisation & modulirisation
- Standardardisation can be done on e.g. component and service level
- Decision is a all levels

Papinniemi, 2019

SUBSYSTEM ENGINEERING PHASE

SUBSYSTEM ENGINEERING INTRODUCTION

INTRODUCTION

- The subsystems and components (items) are fully defined.
- All required information of the system and its subsystem are defined (Note MVP).
- At the end of subsystem engineering, all subsystems are tested and approved.

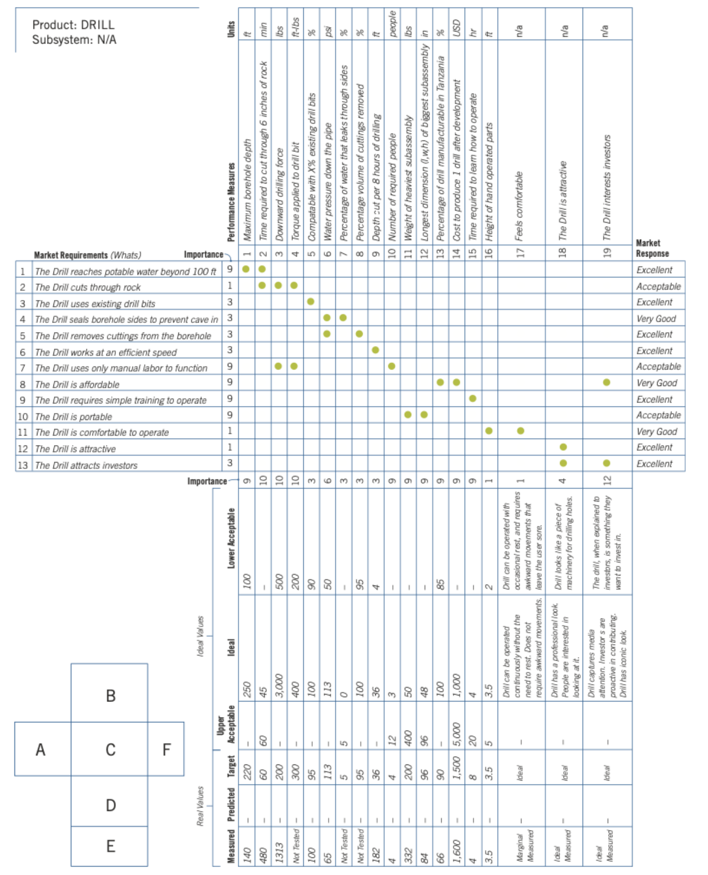
OBJECTIVE OF CONCEPT DEVELOPMENT

- The first version (release) of the product is complete.
- All requirements & tests updated to reflect analysis and testing of subsystems.
- Fully engineered subsystems and components (items) independently verified & approved.
- Manufacturing engineering completed
- Service systems defined (e.g. spares, upgrades)

SUBSYSTEM ENGINEERING INTRODUCTION

- Requirements (system engineering v-curve)
- Tests
- Models
- Prototypes
- Design
- Review and Approval for the design

REQUIREMENT MATRIX



#	Market Requirements (Whats)	Importance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	Market Response
1	The Drill reaches potable water beyond 100 ft	9	●	●																		Excellent
2	The Drill cuts through rock	1		●	●	●																Acceptable
3	The Drill uses existing drill bits	3					●															Excellent
4	The Drill seals borehole sides to prevent cave in	3						●	●													Very Good
5	The Drill removes cuttings from the borehole	3						●		●												Excellent
6	The Drill works at an efficient speed	3									●											Excellent
7	The Drill uses only manual labor to function	9			●	●						●										Acceptable
8	The Drill is affordable	9													●	●					●	Very Good
9	The Drill requires simple training to operate	9															●					Excellent
10	The Drill is portable	9											●	●								Acceptable
11	The Drill is comfortable to operate	1																●	●			Very Good
12	The Drill is attractive	1																		●		Excellent
13	The Drill attracts investors	3																		●	●	Excellent

#	Performance Measures	Units	Importance	Ideal Values: Lower Acceptable	Ideal Values: Ideal	Ideal Values: Upper Acceptable	Ideal Values: Target	Real Values: Predicted	Real Values: Measured
1	Maximum borehole depth	ft	9	100	250	–	220	–	140
2	Time required to cut through 6 inches of rock	min	10	–	45	60	60	–	480
3	Downward drilling force	lbs	10	500	3,000	–	200	–	1313
4	Torque applied to drill bit	ft-lbs	10	200	400	–	300	–	Not Tested
5	Compatable with X% existing drill bits	%	3	90	100	–	95	–	100
6	Water pressure down the pipe	psi	6	50	113	–	113	–	65
7	Percentage of water that leaks through sides	%	3	–	0	5	5	–	Not Tested
8	Percentage volume of cuttings removed	%	3	95	100	–	95	–	Not Tested
9	Depth cut per 8 hours of drilling	ft	3	4	36	–	36	–	182
10	Number of required people	people	9	–	3	12	4	–	4
11	Weight of heaviest subassembly	lbs	9	–	50	400	200	–	332
12	Longest dimension (l,w,h) of biggest subassembly	in	9	–	48	96	96	–	84
13	Percentage of drill manufacturable in Tanzania	%	9	85	100	–	90	–	66
14	Cost to produce 1 drill after development	USD	9	–	1,000	5,000	1,500	–	1,600
15	Time required to learn how to operate	hr	9	–	4	20	8	–	4
16	Height of hand operated parts	ft	1	2	3.5	5	3.5	–	3.5
17	Feels comfortable	n/a	1	Drill can be operated with occasional rest, and requires awkward movements that leave the user sore.	Drill can be operated continuously without the need to rest. Does not require awkward movements.	–	Ideal	–	Marginal Measured
18	The Drill is attractive	n/a	4	Drill looks like a piece of machinery for drilling holes.	Drill has a professional look. People are interested in looking at it.	–	Ideal	–	Ideal Measured
19	The Drill interests investors	n/a	12	The drill, when explained to investors, is something they want to invest in.	Drill captures media attention. Investor s are proactive in contributing. Drill has iconic look.	–	Ideal	–	Ideal Measured

The Requirement Matrix:

- Captures all information about the product in one place
- Translates market requirements into performance measures
- Revised throughout the project

- Six sections of the matrix:
    A. Market Requirements
    B. Performance Measures & Units
    C. Requirement-Measure Correlations
    D. Ideal Values
    E. Real Values
    F. Market Response.

Refer to book Section 11.55

SUBSYSTEM ENGINEERING DESIGN OUTCOMES

REQUIREMENTS

- See system engineering v-curve
- Predicted & measured performance for subsystem
  - Design choices can be made to achieve the required performance (models & Digital Twins)
  - Prototypes of the subsystems are created and tested
  - Results to Section E of a requirements matrix
  - Design changes (ECR/ECO)
- Predicted performance values for the system
  - Measured subsystem performance used to predict system performance
  - Predictions to Section E of a requirements matrix

TEST

- Testing is done continuously in product development
  - Methods & detail will evolve
  - Test result added to the matrix or test systems (e.g. documentation, PLM system)
- Models tested to ***predict***_performance
  - Prediction methods and process must be documented
  - KPI must be predicted
- Prototypes tested to ***measure*** performance
  - Virtual & physical prototypes are used
  - Testing procedure must be documented including test equipment
  - Test result added to the matrix or test systems (e.g. documentation, PLM system)

SUBSYSTEM ENGINEERING DESIGN OUTCOMES

MODELS

- Mathematical models used to predict the behaviour of real systems.
- Values for critical design parameters for modelling are adjusted until the model predictions match target values for the subsystem.
- Engineering models range from handwritten equations to computer-based (Documents vs. Digital Twins).
- Product information & models under revision control.

PROTOTYPES

- Different focus on prototypes to:
    - understand phenomena related to the subsystem
    - measure the performance of the subsystem
- *Testbed* is used to have the values of important design parameters changed quickly
- Information is under revision control & saved to the product definition.

SUBSYSTEM ENGINEERING DESIGN OUTCOMES

DESIGN INFORMATION

- See system engineering v-diagram
- Definition of the system and subsystem had information to:
  - Make or buy the components (items & assy)
  - Assemble component into subsystems
  - Integrate subsystems to system
  - items included in the design follow relevant standards
  - All artifacts under revision control

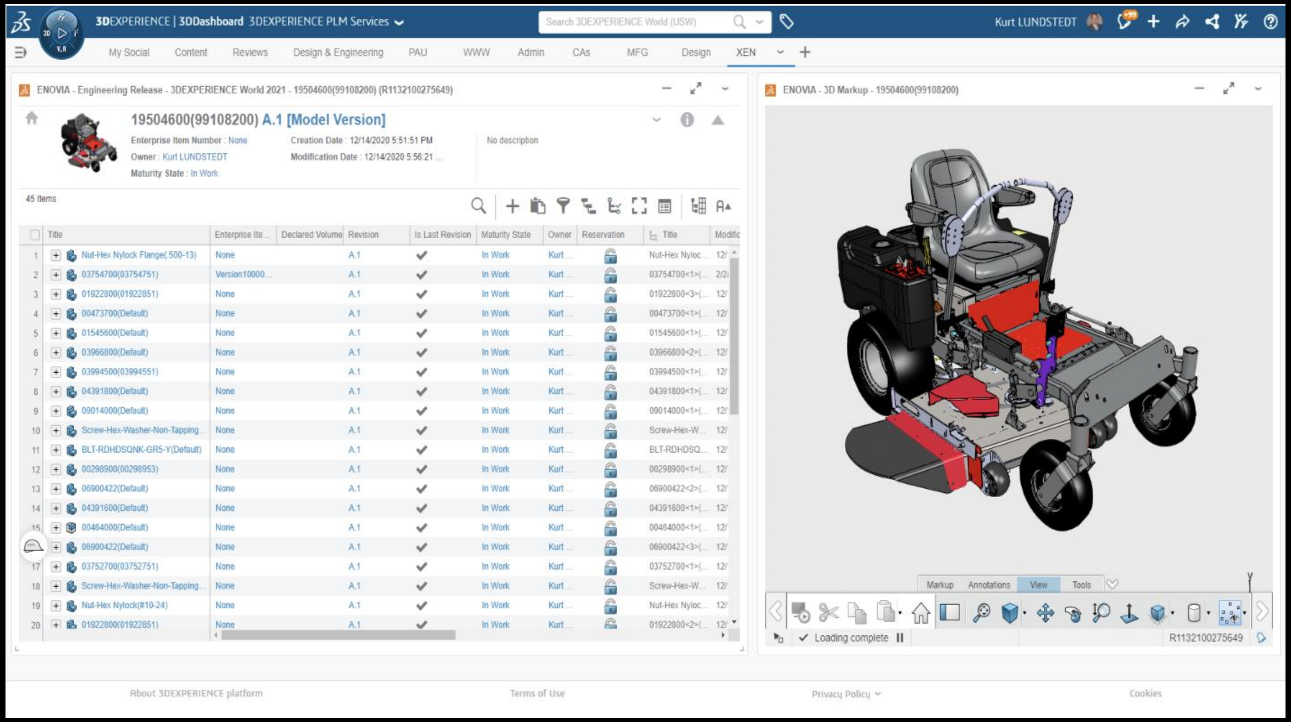
BILL OF MATERIAL (BOM)

- List of all parts in the design in a hierarchical structure:
  - Top assembly, Subassemblies & Parts
  - 3rd part can source all purchased parts and the raw materials for all custom-made parts
  - Build-to-Specification, Build-to-Print , Catalogue items,
- Engineering BOM defines the system and subsystem breakdown
- Manufacturing BOM describes how the system is assembled
- Biil-of-Process defines how the subsystems, assys and items are manufactured or sourced

SUBSYSTEM ENGINEERING DESIGN OUTCOMES

REVIEW & APPROVALS

- All performance measure for the subsystem should have a predicted value and/or a measured value
- Bottom-up approach : items, assemblies, subsystems and system
- Documentation first approvals
- Design definition is under revision control
  - Problem Report (Case Management)
  - Engineering Change Request (ECR)
  - Engineering Change Order (ECO)



PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Requirements	Predicted & measured values for subsystem performance measures.	Section E of the requirements matrix <u>for each subsystem</u>	• All predicted values are present, even if the value is N/A? • All measured values present?	The subsystems meet or exceed the target values of performance. If a few targets are not met, performance is acceptable.
	Predicted values for system performance measures	Section E of the system requirements matrix	• Predicted values for all performance measures? • Consistent with subsystem measured values?	The predicted values meet or exceed the target values for system performance. If a few targets are not met, performance is acceptable.
Tests	Test data demonstrating subsystem performance.	• Reports on methods and results demonstrating the desirability of the subsystem designs. • Plots showing variation of subsystem performance with changes in design parameters.	• Are the test methods complete & correct? • Is there enough detail for a third party to repeat the tests?	Tests carried out with sufficient quality & transferability providing strong evidence for the measured & predicted values.
Models	Engineering models used to choose design parameters & predict values for subsystem performance measures.	• Software source code with run results. • Input files for commercial software with run results. • Excel spreadsheets. MathCAD worksheets. Hand solutions.	• Are the models reported in enough detail to allow a third party to use them?	Not approved directly; used with tests.

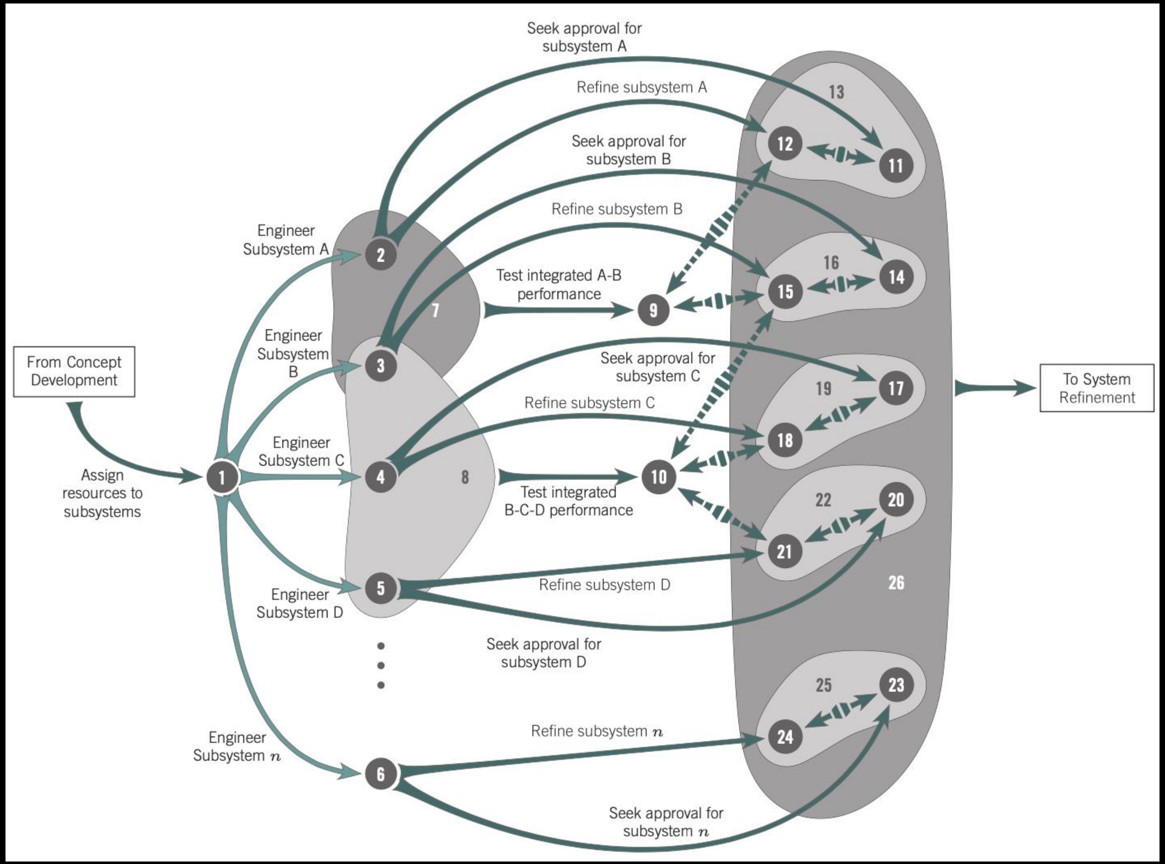
PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Prototype	Prototypes used to help choose values of design parameters. Prototypes used to measure measured values of subsystem performance measures.	• Experimental testbeds used to select design parameter values. • Fully functional subsystems for testing measured values.	Are the prototypes appropriate for the intended use? Is the fidelity and workmanship appropriate?	Not approved directly; used with tests.
Design	Geometric, material, and other appropriate definition of the design for each subsystem.	• Engineering drawings of all custom-designed parts. • Specifications (and possibly ordering information) for all purchased parts. • Assembly drawings for the system and all subsystems. • Schematic diagrams. Piping and/or wiring diagrams. Block diagrams. PC board layout files. • Flowcharts and source code for any software that is part of the design.	Does the design package meet the design intent of the team? Are all relevant standards met? Are all the necessary components included? Is the design package sufficient to allow a third party to correctly make the product and test its compliance with specifications? Are all elements under version control?	The design is sufficiently transferable to allow the creation of complete subsystems and their integration into a complete system by a third party.
	Complete bill of materials (BOM)	• Spreadsheet or database table that lists all known components	Is it complete? Does it have all necessary information? Is it clear and unambiguous?	The bill of materials is appropriate

PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
	Tools:	• Bill of materials • CAD modelling, engineering drawings & checking drawings • design for assembly, design for manufacturing, design of experiments • design structure matrix • dimensional analysis • ergonomics, experimentation • failure modes and effects analysis, fault tree analysis, finite element modelling • prototyping, sensitivity analysis, uncertainty analysis.		
	Common Pitfalls:	• Avoiding analysis • Reinventing analysis • Not doing analysis • Poor experimental procedure (Test Planning & Verification) • Focusing on the prototype rather than the design.		

TOP LEVEL ACTIVITY MAP



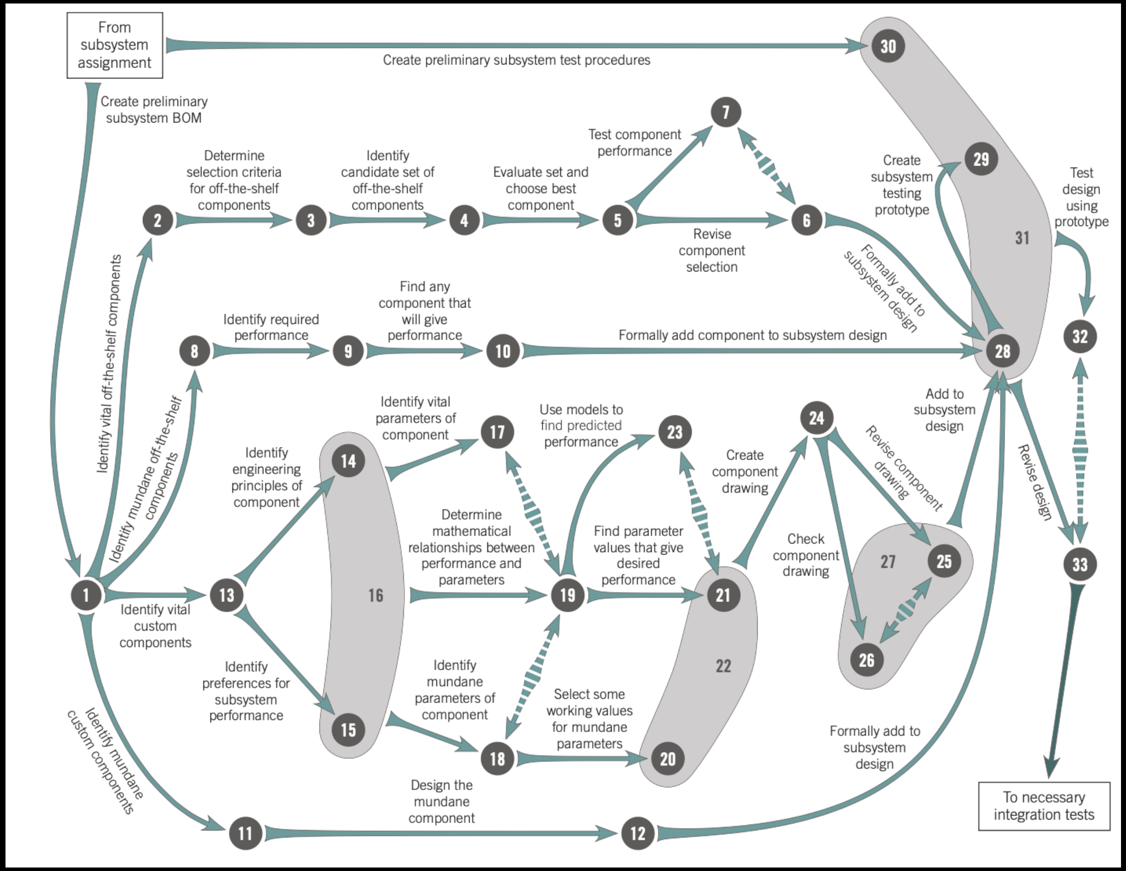
Based on Concept Development Phase

Outcomes from system engineering phase:

- **Sub-system Assignment**
- **Sub-system ready for integration testing (2-6)**
- **Integrated sub-systems (8)**
- **Measured integrated performance (9, 10)**
- **Refined & Approved sub-systems (11-25)**

MOVE TO SYSTEM REFINEMENT

ACTIVITY MAP FOR SUBSYSTEM ENGINEERING



Based on sub-system assignments

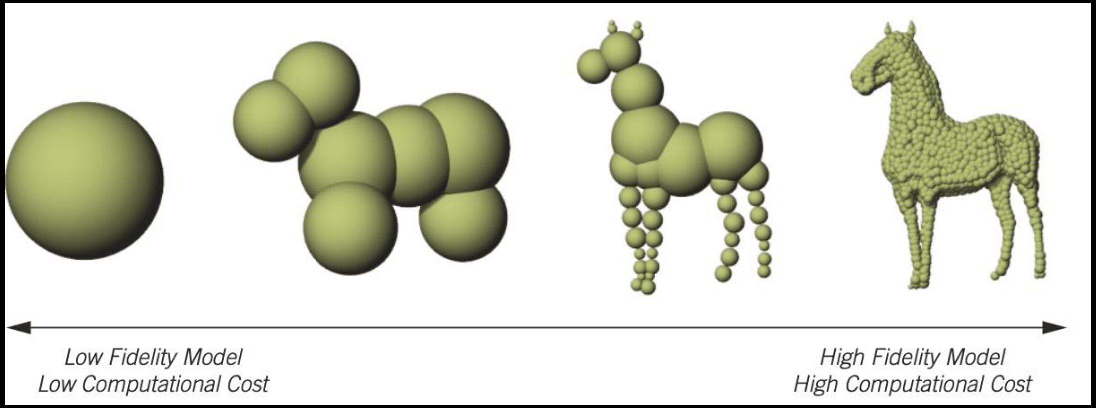
Outcomes from system engineering phase:

- **Identify critical & mundane of the shelf components**
- **Identify critical & mundane custom components**
- **Test design with prototype**
- **Revise design**

MOVE TO INTEGRATION TESTING

USING ENGINEERING MODELS

- Common uses of modelling include the following:
  - Identify the key physics involved in the problem to be solved
  - Explore limits of the design space to help assess the novelty and variety of the candidate solutions
  - Obtain quantitative estimates of the performance of candidate solutions to support the selection of a strong solution candidate
  - Choose appropriate values of design parameters to ensure required performance is obtained
  - Verify that the required performance is obtained.
- Planning for Engineering Models:
  - Define the purpose of the model
  - Define the fidelity (level of approximation)
  - Define the experimental plan for the model
  - Define the schedule for the modelling



	"Back of the Envelope" Model	Fundamental Engineering Model	Discretized Model	Physical Model
Symbolic solution available	●	●		
Numerical solution available, but not symbolic		●	●	
Experimental solution available, but not computational				●

USING ENGINEERING MODELS

Simplified Model

- "Back-of-the-envelope" models
- Based on fundamental scientific principles & approximations

Fundamental Engineering Model

- Based on fundamental scientific principles
- More refined approximations.

Physical Model

- Based on experimental results.
- Used when unclear how to create other model types of the system
- Uncertainty how physical principles apply.

Discretized model (FEM)

- Based on models applied to discrete portions of the system,
- System performance calculated from discrete portions.
- Discretized models are used when the geometry or boundary conditions complex to create a fundamental engineering model.
- The system is discretized, and appropriate numerical models for each of the discrete elements are created, with consistent boundary conditions between the elements. All of the discrete models are solved, which leads to a numerical solution for the entire system.

PROTOTYPES AND THEIR PURPOSE

Purpose	Description
Learning	• Understand how the design ideas work in the physical world. • Related to physical engineering models.
Communication	• Share the current state of design. • Allows people to obtain sensory input about the product.
Integration	• See how well the parts of the design work together. • The performance testing & validation prototypes at the end of the system refinement stage are examples of integration prototypes, but other partial integration prototypes can be effectively used in product development.
Milestone	• Provide a concrete goal for the team to work toward. • A milestone prototype can be a very effective way of getting the development team to finalize the design by a milestone deadline.
Performance Testing	• Provide a means for the development team to demonstrate that the performance measures (both subjective and objective) are being met.
Validation	• Provide a means for market representatives to express their perception of how well the product meets market requirements.


ANY
QUESTIONS
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